

# Educational Attainment and Unemployment in Pennsylvania, 2020-2024

## Abstract

This report explores the relationship between education level and unemployment rate among the non-military workforce aged 25 to 64 in Pennsylvania. It aims to explore whether higher levels of education are associated with lower unemployment after controlling variables such as age, sex, race and year. We compared the weighted unemployment rates of different education levels and used the weighted logistic regression model to analyze these relationships. The research results show that there is a correlation between the two, but it is not a causal relationship.

## Data, sample, and feature selection

The data comes from the U.S. Census Bureau's 2020-2024 ACS five-year PUMS Pennsylvania personal data set. The analysis subjects included 249,839 adults aged 25 to 64 who belonged to the civilian labor force, including 9,429 unemployed respondents. The unemployment status code is 1 and the employment status code is 0.<sup>1</sup>

Education (SCHL) is grouped into five levels: less than high school, high school/GED, some college/associate, bachelor's, and graduate degree. Bachelor's degree is the reference category. Age, sex, race, and year labels are included. Person weights (PWGTP) are used in descriptive statistics and regression.<sup>2</sup>

Table 1. Sample characteristics (weighted where indicated)

| Characteristic            | Value   |
|---------------------------|---------|
| Observations (unweighted) | 249,839 |
| Mean age (weighted)       | 43.73   |
| Female (weighted)         | 47.65%  |
| Less than high school     | 5.17%   |
| High school / GED         | 27.33%  |
| Some college / associate  | 25.23%  |
| Bachelor's degree         | 25.40%  |
| Graduate degree           | 16.87%  |
| Overall unemployment rate | 4.45%   |

## Exploratory patterns and trends

The data show that the weighted unemployment rate tends to decline as educational attainment increases. Specifically, the unemployment rate is 8.15% for those who did not complete high school, 6.01% for those with a high school diploma or GED, 5.05% for those with some college or an associate's degree, 3.11% for those with a bachelor's degree, and 1.94% for those with a graduate degree. The gap in unemployment rates between the group with the lowest level of education and the group with the highest is 6.21 percentage points. However, other factors, such as occupation and work experience, may also influence these figures.

<sup>1</sup> U.S. Census Bureau. American Community Survey (ACS) 2020–2024 5-Year PUMS, Pennsylvania person file. [https://www2.census.gov/programs-surveys/acs/data/pums/2024/5-Year/csv\\_ppa.zip](https://www2.census.gov/programs-surveys/acs/data/pums/2024/5-Year/csv_ppa.zip)

<sup>2</sup> U.S. Census Bureau. 2024 ACS PUMS Data Dictionary and Code Lists, variables AGEP, SCHL, ESR, SEX, RAC1P, and PWGTP. <https://www.census.gov/programs-surveys/acs/microdata/documentation.html>

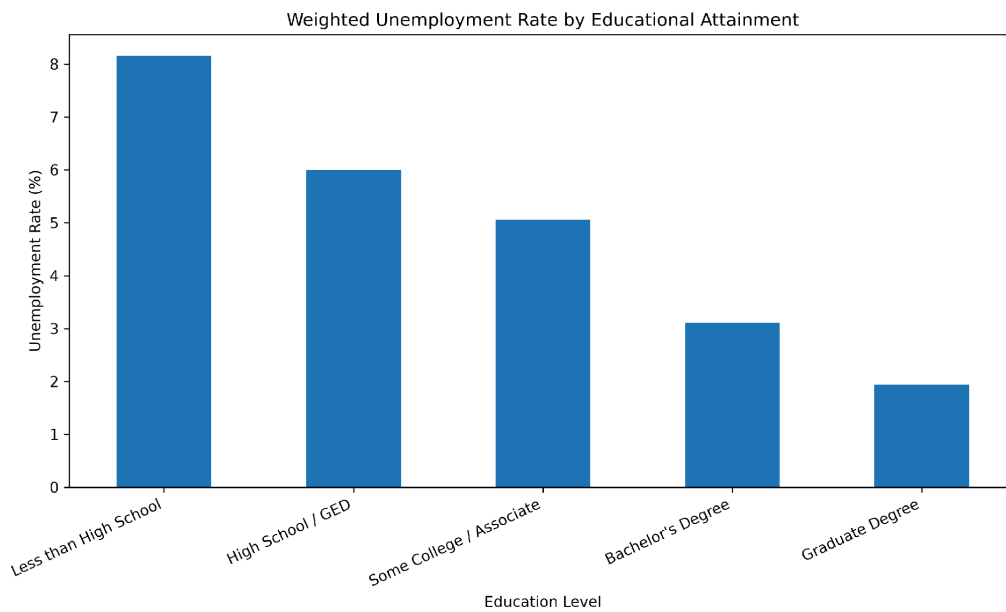


Figure 1. Survey-weighted unemployment rate by educational attainment.

### Changes across year labels

The unemployment rate also varies across year labels used in the notebook. For adults with less education than high school, the unemployment rate has changed from 9.89% in 2020 to 11.65% in 2021, 4.70% in 2022 and 8.07% in 2024. We can see that the unemployment rate of postgraduate degree holders has always been low. However, since these year labels are derived from the SERIALNO field in a summarized five-year ACS dataset, these trends should be carefully interpreted.

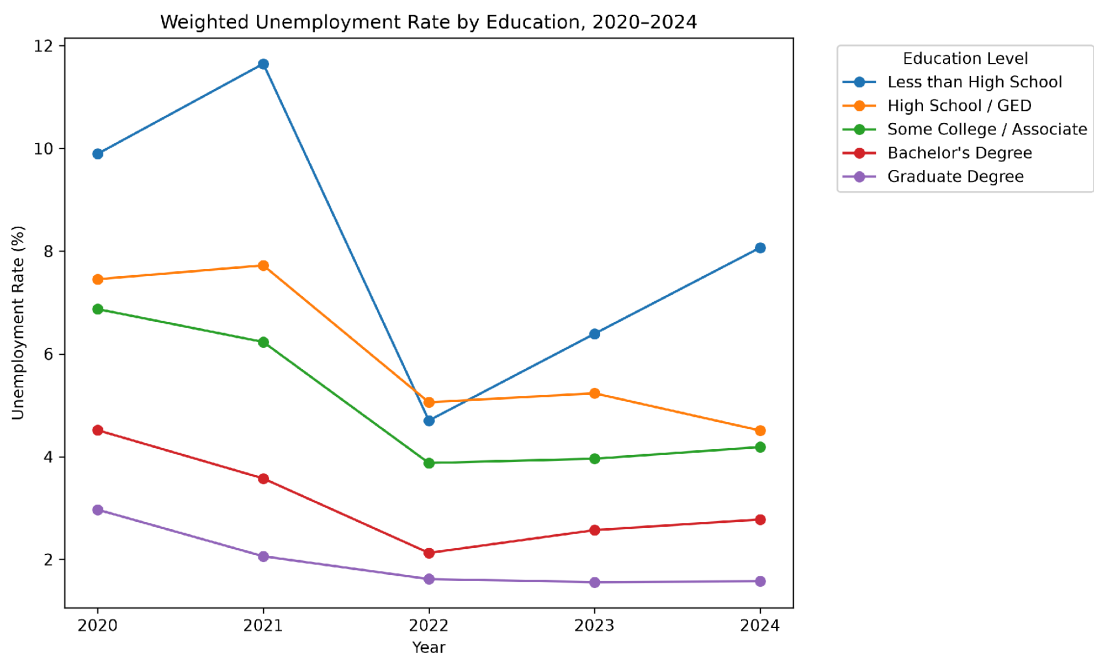


Figure 2. Weighted unemployment rates by education and notebook-derived year label; not independently validated annual estimates.

### Model development and statistical results

The analysis utilized a binomial generalized linear model with a logit link function, using educational attainment, age, gender, race, and year as predictor variables to estimate the probability of unemployment. This analysis used PWGTP as the frequency weight. With all included variables held constant, this ratio summarizes the difference relative to college graduates. For adults without a high school diploma, the estimated odds ratio was 2.565 (95% confidence interval: 2.521–2.610). The odds ratio for high school/GED diploma holders was 1.958. Holders of graduate degrees had a relatively low unemployment odds ratio (0.636). It is important to note that these are comparisons of odds ratios, not percentage changes in the probability of unemployment.<sup>3</sup>

Table 2. Adjusted education odds ratios (reference: bachelor's degree)

| Education                | Odds ratio | 95% CI      |
|--------------------------|------------|-------------|
| Less than high school    | 2.565      | 2.521-2.610 |
| High school / GED        | 1.958      | 1.935-1.982 |
| Some college / associate | 1.585      | 1.566-1.605 |
| Graduate degree          | 0.636      | 0.624-0.647 |

Note: All four education coefficients have reported p-values below 0.001 in the fitted output; intervals and p-values use the notebook's default nonrobust covariance estimates.

### Adjusted predictions and model interpretation

After controlling the influence of age, gender, race and year through the model, we can see a clear pattern: higher education is associated with lower risk of unemployment. Specifically, the adjusted forecast unemployment rate has dropped from 7.67% of 'less than high school' to 2.04% in the graduate group through high school/GED (5.98%), some college/associate (4.91%) and bachelor's degree (3.16%). It should be noted that this law reflects the close correlation between the two, not the causal conclusion that the improvement of education directly leads to a decrease in unemployment.<sup>4</sup>

---

<sup>3</sup> statsmodels developers. Generalized Linear Models (GLM), binomial family, logit link, and frequency weights (software documentation). <https://www.statsmodels.org/stable/glm.html>

<sup>4</sup> Project team. Final\_Project\_Analysis.ipynb, adjusted predictions and PWGTP-weighted averages. [https://github.com/linyuz-0419/90819\\_FinalProject](https://github.com/linyuz-0419/90819_FinalProject)

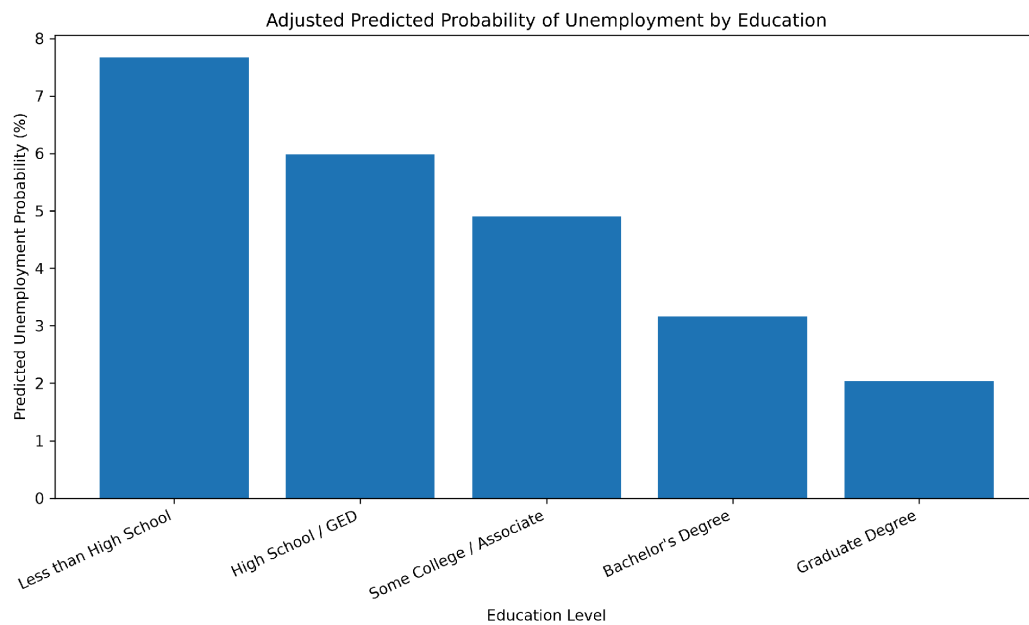


Figure 3. Average model-adjusted predicted unemployment probability by education.

### Evaluation and validation

The model extracts 20% of the test data to test the effect of the model. Although the final accuracy rate seems to be 96.2%, it is mainly because unemployment itself is a very small probability event; in fact, the model predicts that everyone is employed, and the precision and recall for unemployed are all 0. In addition, the ROC-AUC of 0.659 is also relatively limited, which means that it is difficult to accurately distinguish who will be unemployed. Therefore, the real value of this model is to help us clarify the statistical relationship between education and employment, not to predict the unemployment fate of individuals.<sup>5</sup>

### Limitations

There are several points that need to be noted in the explanation of this study: first, in addition to education, a person's working years and any industry will directly affect employment. The model cannot fully control these factors, so it can only show that it is related, and cannot prove that higher education directly leads to less unemployment. Second, we use 5-year consolidated data, and the forced disassembly of a single year according to the number does not represent the real independent trend of each year. Finally, the model has not fully taken care of the complex sampling characteristics of the census data in terms of statistics, and because the number of unemployed is relatively small, it is not accurate to use it to predict which person will be unemployed.<sup>6</sup>

### Data and reproducibility

Source: U.S. Census Bureau, 2020-2024 ACS five-year PUMS (Pennsylvania person file). Analysis and figures: Final\_Project\_Analysis.ipynb; prepared data: pa\_cleaned.csv. Repository: [https://github.com/linyuz-0419/90819\\_FinalProject](https://github.com/linyuz-0419/90819_FinalProject)

<sup>5</sup> scikit-learn developers. Model evaluation: precision, recall, confusion matrix, and ROC AUC (software documentation). [https://scikit-learn.org/stable/modules/model\\_evaluation.html](https://scikit-learn.org/stable/modules/model_evaluation.html)

<sup>6</sup> U.S. Census Bureau. 2020–2024 ACS 5-Year PUMS Accuracy. <https://www.census.gov/programs-surveys/acs/microdata/documentation/2024.html>